

Samples of Exam Questions

This document contains sample exam questions in the same style as those on the final exam of the course. You will find attached all the files (materials and datasets) needed to complete the "numerical" exercises. The following approach to scoring responses is implemented and is based on "One best answer":

- There is always only one correct answer – a response that is more correct than the rest.
- Students are only able to select one answer per question.
- The score of each question will be given next to the question number (not here, but it will in the real exam).
- Every incorrect answer corresponds to 0 points (incorrect answers do not result in subtraction of points).

Question 1:

Consider the RGB color space. Are the following statements correct or wrong?

Select the correct answers

	Correct	Wrong
The RGB colour space can describe grey shades	☒	☐
The RGB colour space encodes light intensity explicitly	☐	☒
The RGB colour space defines each color based on 3 values	☒	☐

Question 2:

Can we perform corelation using convolution?

Choose one answer

- ☒ Yes, by modifying the kernel
- ☐ Yes, the outcome of corelation and convolution are always the same
- ☐ No, correlation and convolution are fundamentally different
- ☐ No, we can only perform convolution using correlation, but not the inverse

Question 3:

Choose one answer

- ☐ The Harris detector uses the Sobel edge detection
- ☐ The Harris detector uses the Canny edge detector
- ☒ The original Harris detector is uses a $\begin{bmatrix} -2 & -1 & 0 & 1 & 2 \end{bmatrix}$ filter

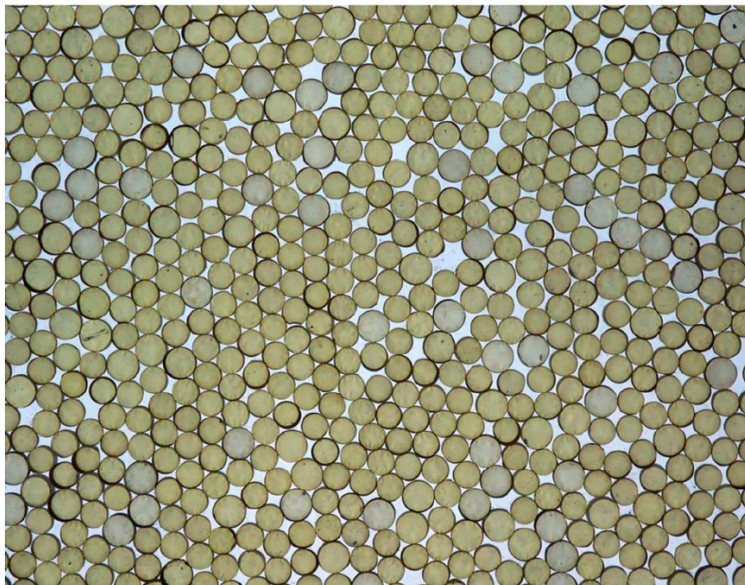
Question 4:

Using the following image (in the files you downloaded: images/circles.jpg), do the following

1. Convert it to grayscale (do not change the size!).
2. Apply Canny edge detection (cv2.Canny) with the parameters
 - threshold1=100
 - threshold2=200
3. Detect circles on the output of the Canny detection. Use cv2.HoughCircles with the following parameters
 - method=cv2.HOUGH_GRADIENT
 - dp=1
 - minDist=20
 - param1=50
 - param2=30
 - minRadius=10
 - maxRadius=50

How many circles are detected with these parameters? Choose the answer that is closest to your result.

NB: The number is not equal to the total count of circles in the image, but rather what the detector detects.



Choose one answer

- ☐ 180
- ☐ 769
- ☒ 313
- ☐ 550

Question 5:

Which surface is more problematic for a stereo correspondence algorithm?

Choose one answer

- ☒ A flat black wall.
- ☐ A concrete wall with a graffiti.
- ☐ A old stone wall.

Question 6:

Apply Linear Regression to the provided data to obtain a model of the form $y=ax+b$.
Follow the provided guidelines for the implementation.

What are the values of the parameters "a" and "b" in that model (APPROXIMATELY)?

Choose one answer

- ☐ $a=0.23, b=25.60$
- ☒ $a=0.55, b=19.86$
- ☐ $a=0.82, b=19.86$
- ☐ $a=0.55, b=12.37$
- ☐ $a=0.23, b=12.37$
- ☐ $a=0.82, b=25.60$